

113A Introduction to Computer and Computer Science

Final Project

Due: 12/23 12:00:00

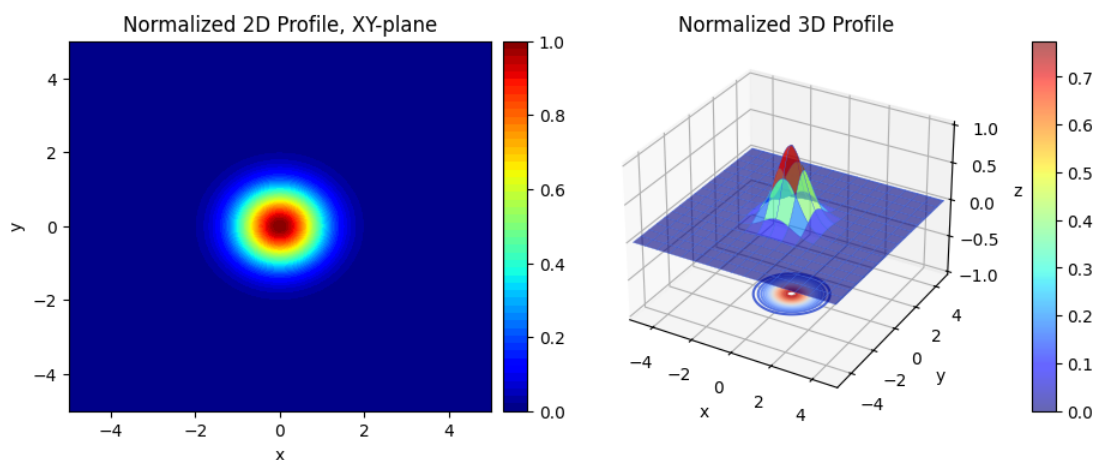
Laser is one of the most important technologies in modern optics. Since light can be regarded as an electromagnetic radiation, we can use an equation to describe its electric field. Although you have not learned any related knowledge about this topic, we can still write some program to plot its profile in the transverse plane. Don't worry, I will give you the equation.

Problem #1: Gaussian Beam (30%)

A Gaussian beam is an idealized beam of electromagnetic radiation whose amplitude envelope in the transverse plane is given by a Gaussian function.

$$E(x, y) = \exp[-(x^2 + y^2)]$$

Please write a program to plot its 2D and 3D profiles.



FRIENDLY REMINDER

You don't need to plot a figure that is exactly same as mine.

Problem #2: Higher-order Modes (30%)

As a special case of electromagnetic radiation, Gaussian beams are solutions to the wave equation for an electromagnetic field in free space or in a homogeneous dielectric medium. This means Gaussian beam is not the only solution of wave equation. There are other solutions called “higher-order modes.” We can use Hermite polynomial to describe the profile of these higher-order modes.

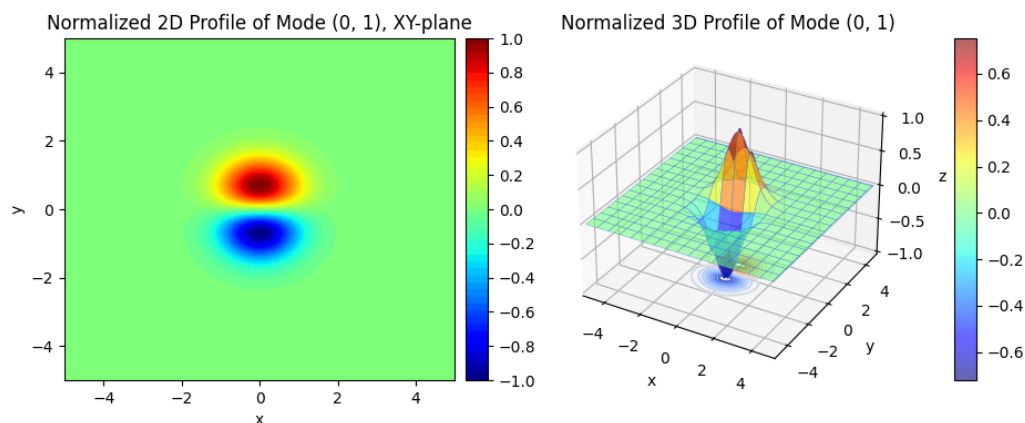
$$E_{l,m}(x, y) = H_l(\sqrt{2}x)H_m(\sqrt{2}y) \exp[-(x^2 + y^2)]$$

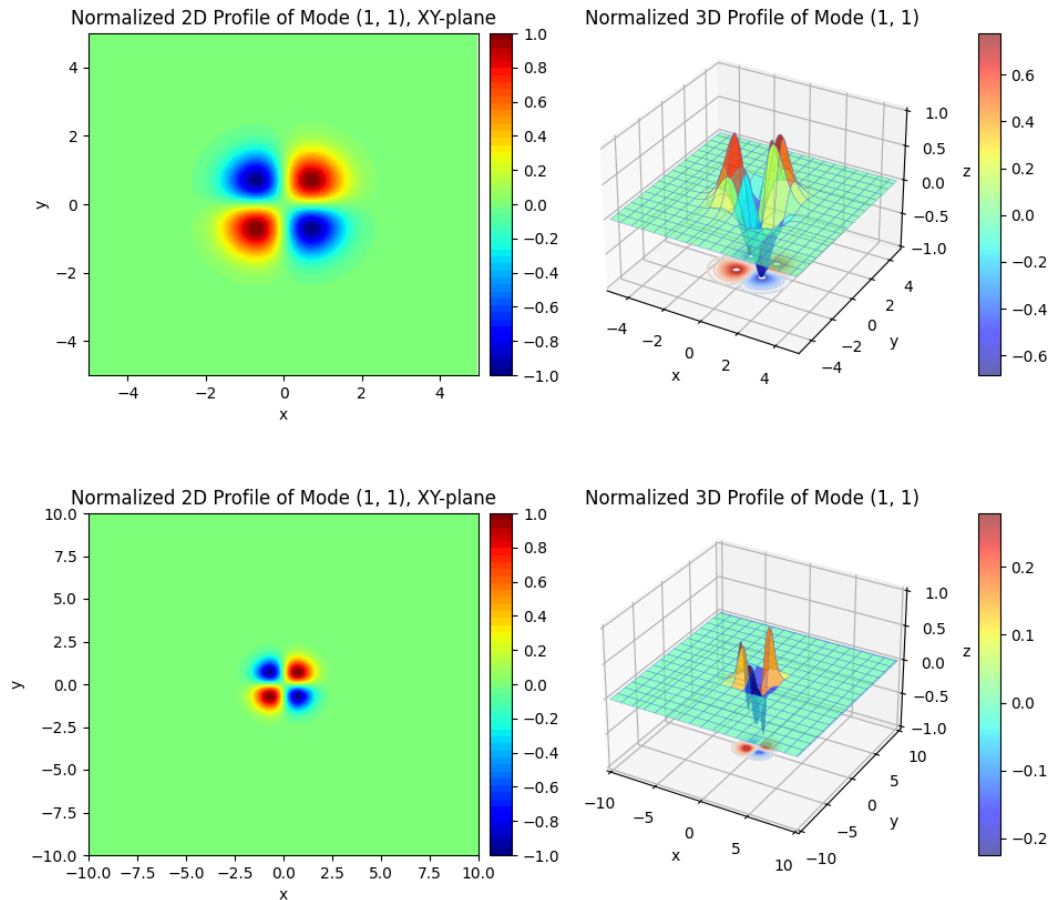
The first 7 Hermite polynomials are:

Index	Equation
0	$H_0(x) = 1$
1	$H_1(x) = 2x$
2	$H_2(x) = 4x^2 - 2$
3	$H_3(x) = 8x^3 - 12x$
4	$H_4(x) = 16x^4 - 48x^2 + 12$
5	$H_5(x) = 32x^5 - 160x^3 + 120x$
6	$H_6(x) = 64x^6 - 480x^4 + 720x^2 - 120$

Please write a program to plot the 2D and 3D profiles of higher-order mode. This program should allow user to choose a specific mode (l, m) , number of points along both axes, and the side length. For example, you might get these results.

```
(nycudopcs) 0:\ >python main_final.py --mode 0 1
(nycudopcs) 0:\ >python main_final.py --mode 1 1 --num 200
(nycudopcs) 0:\ >python main_final.py --mode 1 1 --len 10
```





Please just provide the same information and feel free to change the style.

Problem #3: Summarize your final project (40%)

Please write a report for this project. You should explain the logic of your program and why you choose that mode to plot. Here is a check list for your report:

Index	Item	Check
1	Your name (otherwise you will get 0)	☐
2	Name of collaborator(s)	☐
3	Your result (figures)	☐
4	Why do you choose this mode?	☐
5	Explain the logic of your program	☐
6	Anything you would like to say in the end of this semester	☐

Note. You are welcome to add additional content in your report.

Your report should be a .pdf file.

Here is the sample code:

```
from func import solution
import argparse

if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="2024 NYCUDOPCS Final")
    parser.add_argument("--num", default=500, type=int, help="Number of points (default: 500)")
    parser.add_argument("--len", default=5, type=float, help="Half side length (default: 5)")
    parser.add_argument("--mode", default=(0, 0), type=int, nargs="+", help="TEM mode (default: (1, m) = (0, 0))")

    opt = parser.parse_args()

    for value in opt.mode:
        if value > 6 or value < 0:
            raise ValueError("Mode should be an integer between 0 to 6.")

    if opt.len <= 0:
        raise ValueError("Half side length should be a positive number.")

    if opt.num <= 1:
        raise ValueError("The number of points should be an integer larger than 1.")

    solution(config=opt)
```

Please accomplish this homework with an organized code (e.g., with main script and function script). **Your main script file name must be ‘main_final.py.’** Here is a template for your code structure:

```
113A_final_0123456789
├─ report.pdf      # Report of your final project
├─ module1.py      # Module 1 for final
├─ module2.py      # Module 2 for final
├─ :               # Other modules if necessary
└─ main_final.py   # Main script of final
```

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs. Here is the template:

```
"""
Created on Wed Aug 7 01:23:45 2024

@author: Xi Winnie, student ID

@collaborators: William Lai, his student ID
"""
```

Please save your homework as a “.zip”, “.7z”, or “.rar” file, where the file name should follow this format:

113A_final_ID.zip

For example,

113A_final_0123456789.zip

Please note. **We are not going to accept any homework file with wrong file name, wrong file format, or without signature in main script.** Please double check the content of your file.

Once you have accomplished your works, you can submit your homework to the “E3@NYCU” system. There will be a section for uploading your homework.